

Comunicazione Cellulare

In biology, **cell communication** is the process by which a cell interacts with itself, with other cells, and/or with the environment.

Cell signaling can occur over short or long distances:

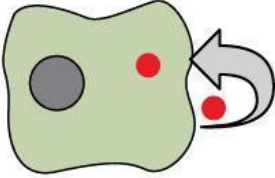
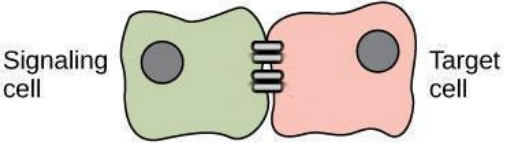
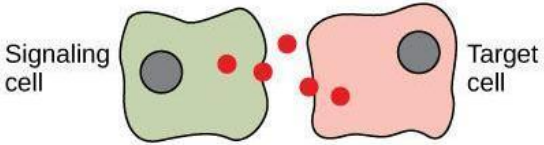
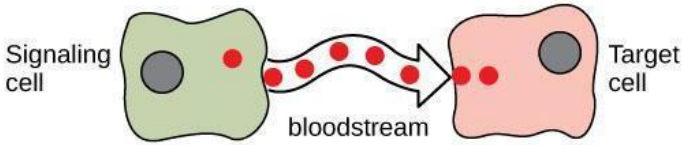
Autocrine signaling occurs when the signal acts on the same cell generating the signal.

Juxtacrine (contact dependent) signaling occurs between physically adjacent cells.

Paracrine signaling occurs between nearby cells.

Endocrine signaling occurs between distant cells.

Typically, the signaling process involves three components: the **signal**, the **receptor**, and the **effector(s)**.

Forms of Chemical Signaling	
Autocrine	A cell targets itself.
	
Contact-Dependent	Cells must physically interact to initiate signaling
	
Paracrine	A cell targets a nearby cell.
	
Endocrine	A cell targets a distant cell through the bloodstream.
	

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Signal

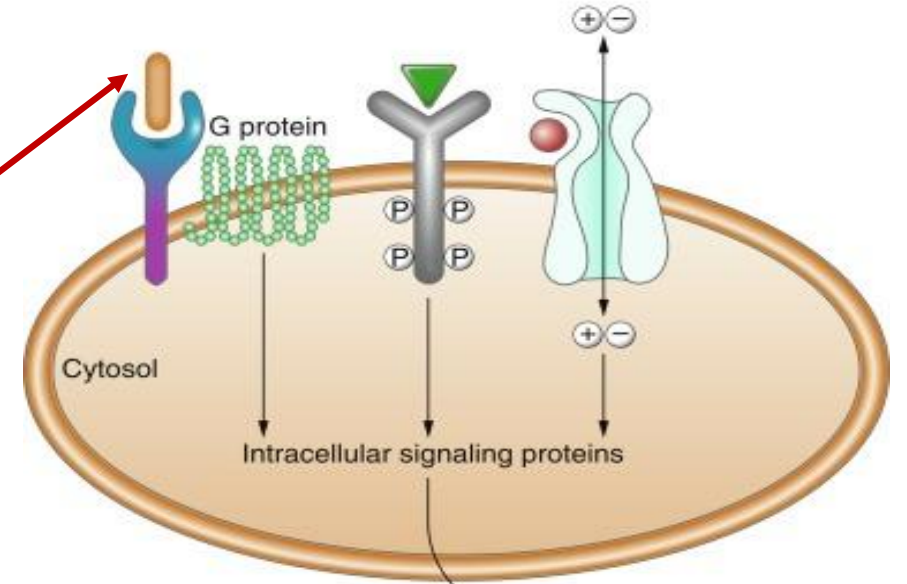
Physical signals can be pressure, voltage, temperature, or light.

Chemical signals include ions, lipids, carbohydrates, glycosylated proteins, peptides, nucleic acids.

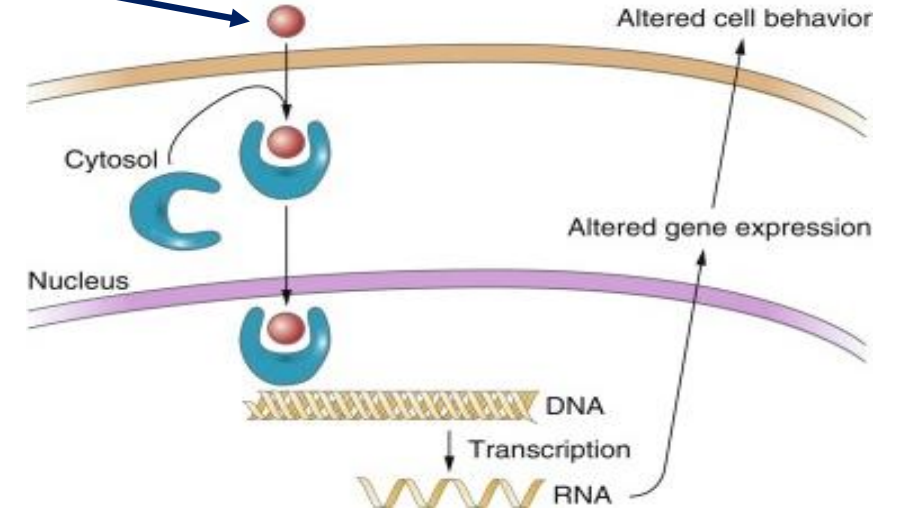
Hydrophilic molecules are unable to diffuse freely across the lipid bilayer of the plasma membrane, so their action is mediated by a cell membrane bound receptor.

Lipophilic molecules can diffuse passively across the plasma membrane and interact with intracellular receptors.

A Cell surface receptor



B Intracellular receptor



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Receptor

Receptors are single proteins or complex of proteins, located in the plasma membrane or in the cytoplasm, organelles, and nucleus having the ability to detect a signal either by binding the specific chemical or dealing with physical agents.

Membrane Receptors

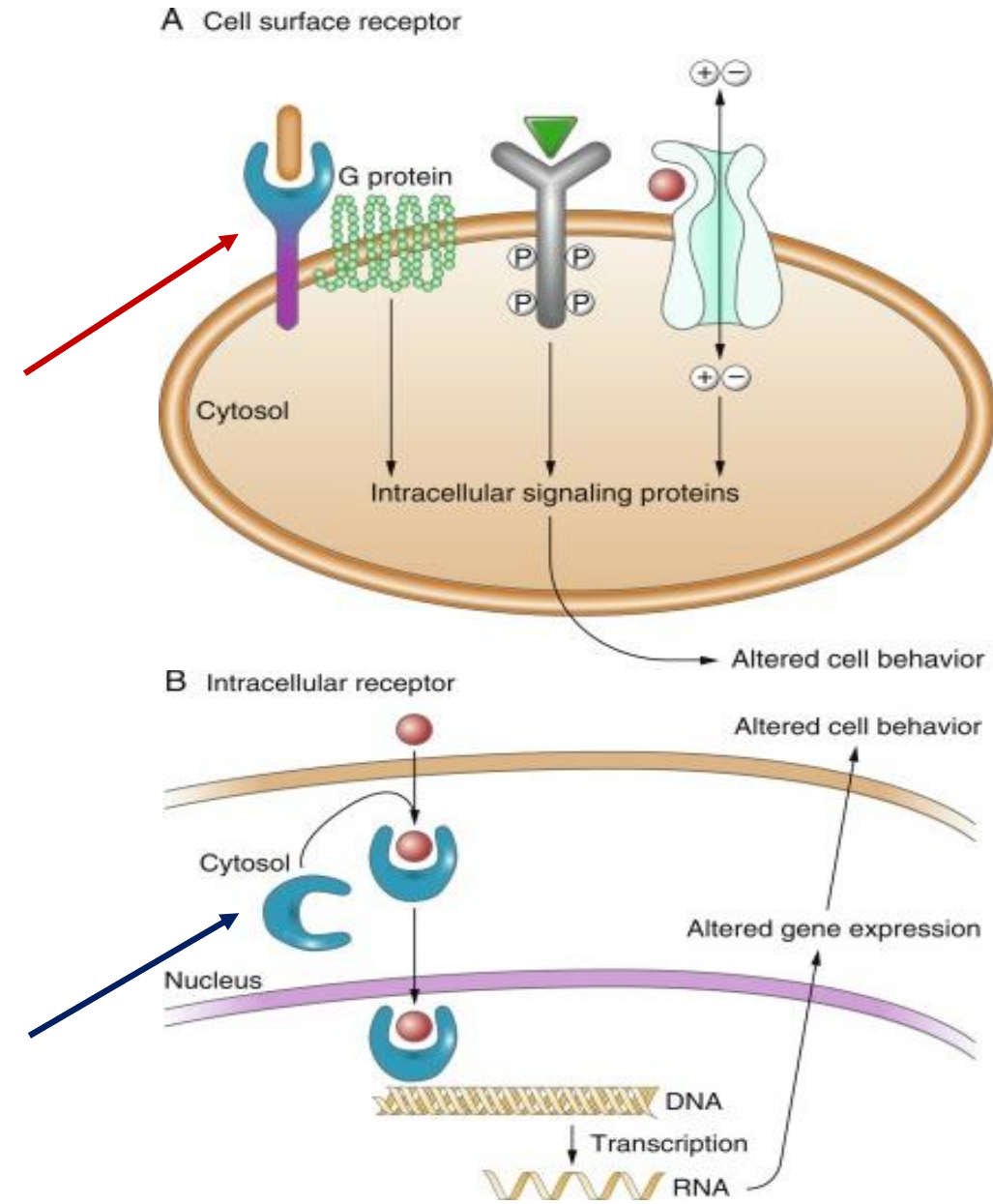
Membrane Receptors are proteins embedded within the plasma membrane. The extracellular domain is responsible for the interaction with a specific ligand. The intracellular domain is responsible for the initiation of a cascade of chemical reactions which ultimately triggers the specific cellular function controlled by the receptor.

Ion channels receptors are large transmembrane proteins with a ligand activated gate function. When these receptors are activated, they may allow or block passage of specific ions across the cell membrane. Most receptors activated by physical stimuli such as pressure or temperature belongs to this category.

Enzyme-linked receptors are characterized by an intracellular domain with enzymatic or catalytic activity. Upon activation the enzymatic portion is responsible for promoting specific intracellular chemical reactions.

Intracellular Receptors

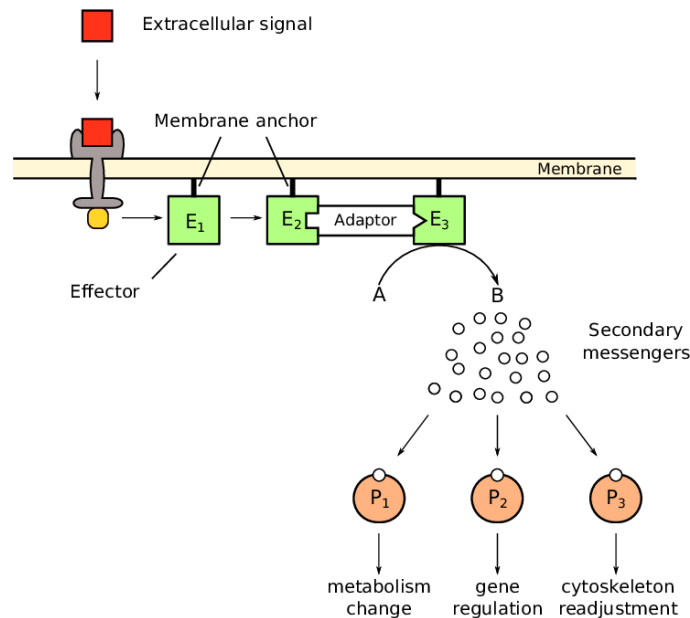
Intracellular receptors are generally located in the cytoplasm. They can bind to lipid soluble ligands that diffuse passively through the plasma membrane such as steroid hormones or lipophilic molecules freely circulating in the cytoplasm.



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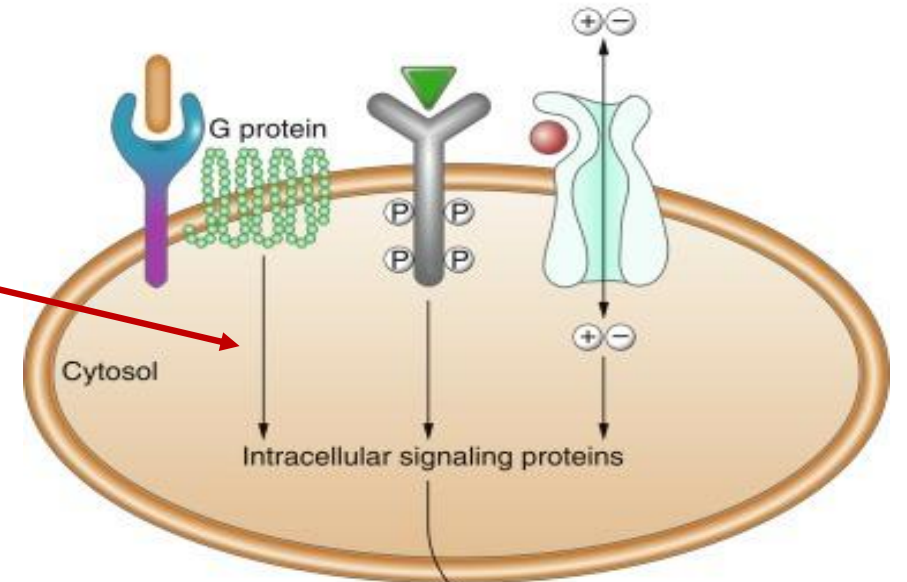
Effectors

Typically, the final effect can consist in the change of ion concentration across the membrane, or the post-translational modification of down-stream proteins leading to activation/inhibition of precise molecular pathways, or the shuttling of the hormone-receptor complex inside the nucleus and the transcriptional modulation of specific targeted genes.

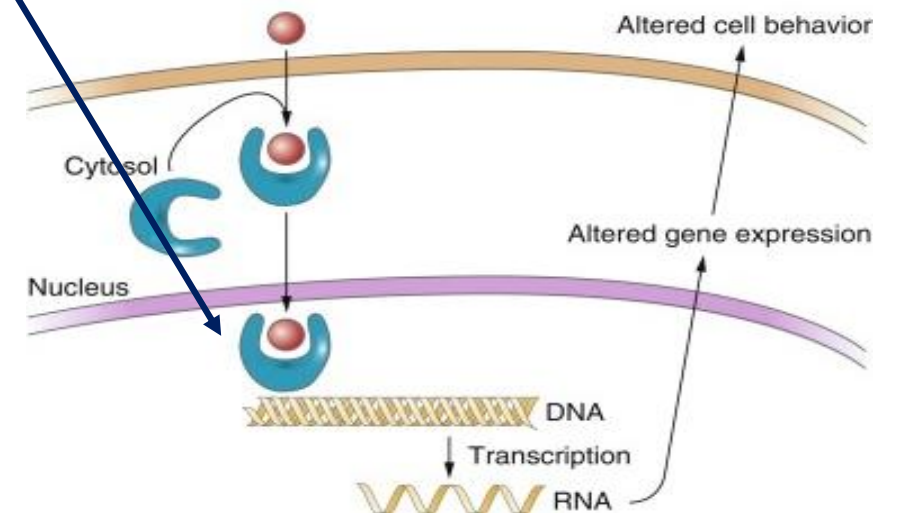


The signal, by interacting with the receptor, starts a series of molecular events within the cell (signal transduction) leading to the final effect. The initiation of a second messenger system cascade propagating the signal through the cell can amplify the initial signal (the first messenger).

A Cell surface receptor



B Intracellular receptor



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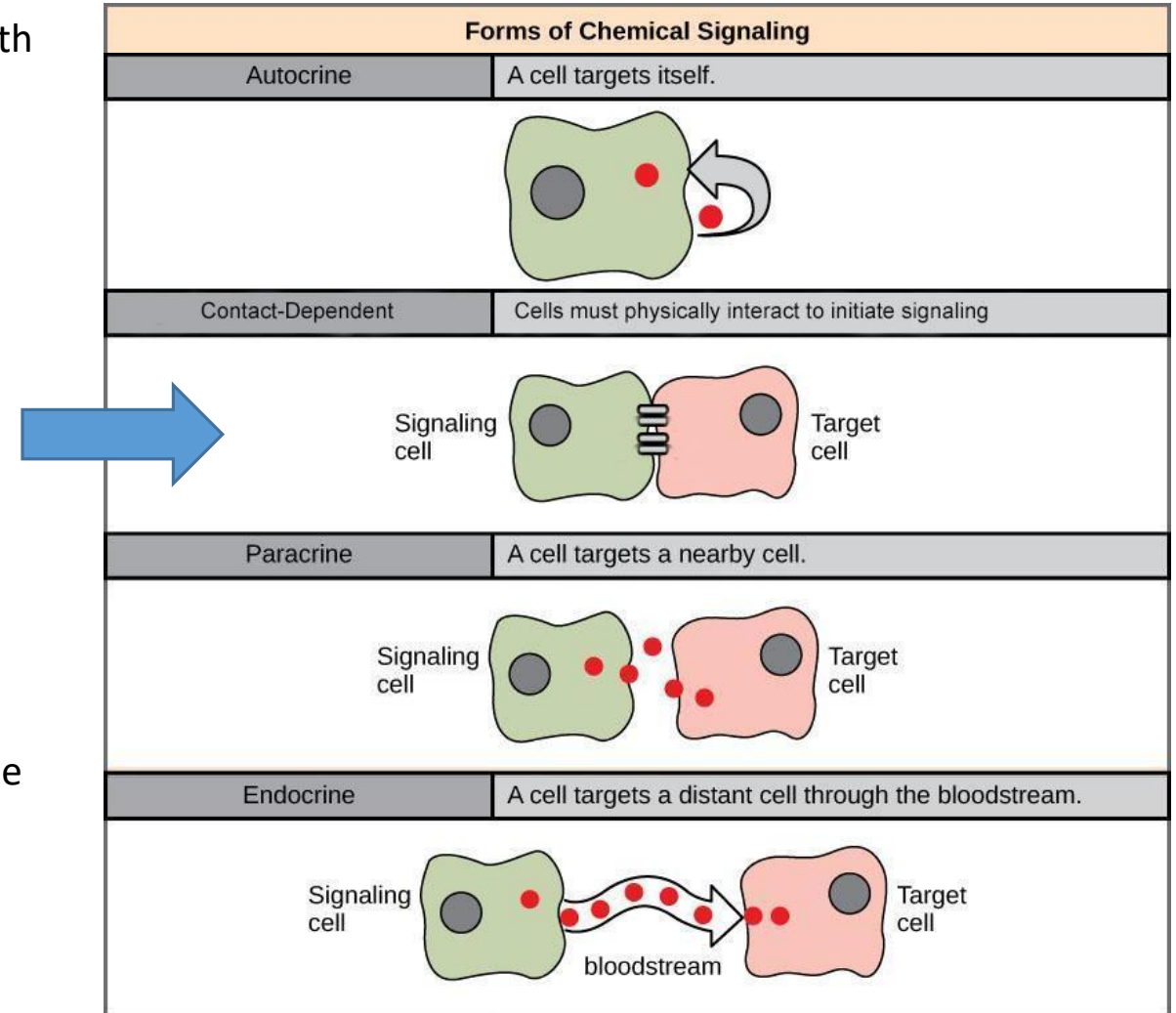
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T-Cell Immunity

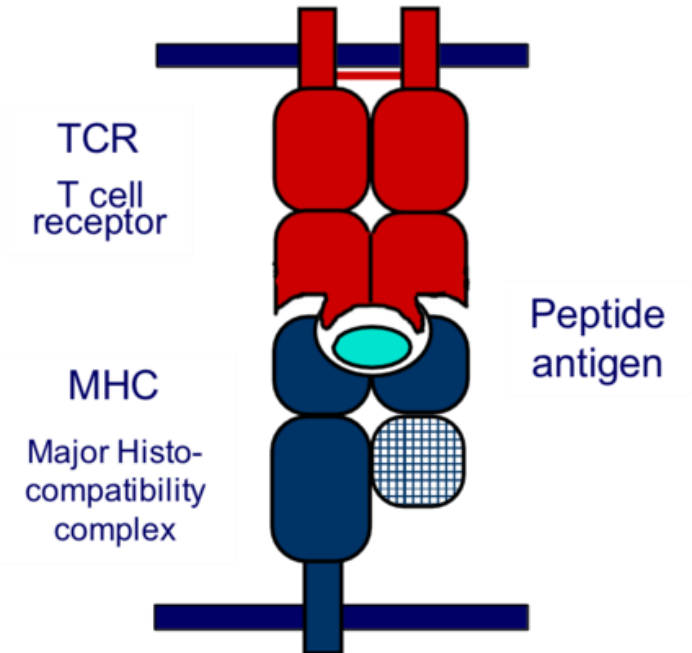
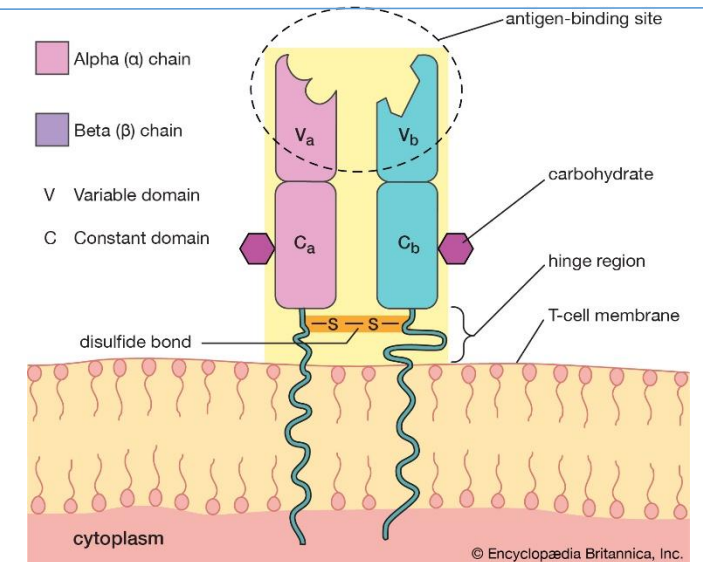
T cells play a central role in cell-mediated immunity. By expressing the T-cell receptor (TCR) on the cell surface they recognize self and non-self antigens. **Genomic rearrangements of TCR gene units through a process called V(d)J, generates the stochastic variability in antigen recognition domain of TCR essential for a functional immune surveillance.**

Cytotoxic T cells (CD8+ TC cells) recognize their targets by binding to antigens associated with **MHC class I** molecules. TC cells release the cytotoxins perforin, granzymes, and granulysin. Through the action of perforin, granzymes enter the cytoplasm of the target cell and their serine protease function triggers the caspase cascade leading to apoptosis.

T helper cells (CD4+ TH cells) assist other white blood cells in immunologic processes, including maturation of B cells into plasma cells and memory B cells, activation of cytotoxic T cells and macrophages. TH cells recognize their targets by binding to antigen associated with **MHC class II** molecules. Once activated, they divide rapidly and secrete cytokines that regulate or assist other immune cells "effector cells" in the active immune response.

Memory T cells (CD4+ or CD8+) are a subset of antigen-specific T cells that persist after the immune response. They quickly expand to large numbers of effector T cells upon re-exposure to their cognate antigen, thus providing the immune system with "memory" against past infections.

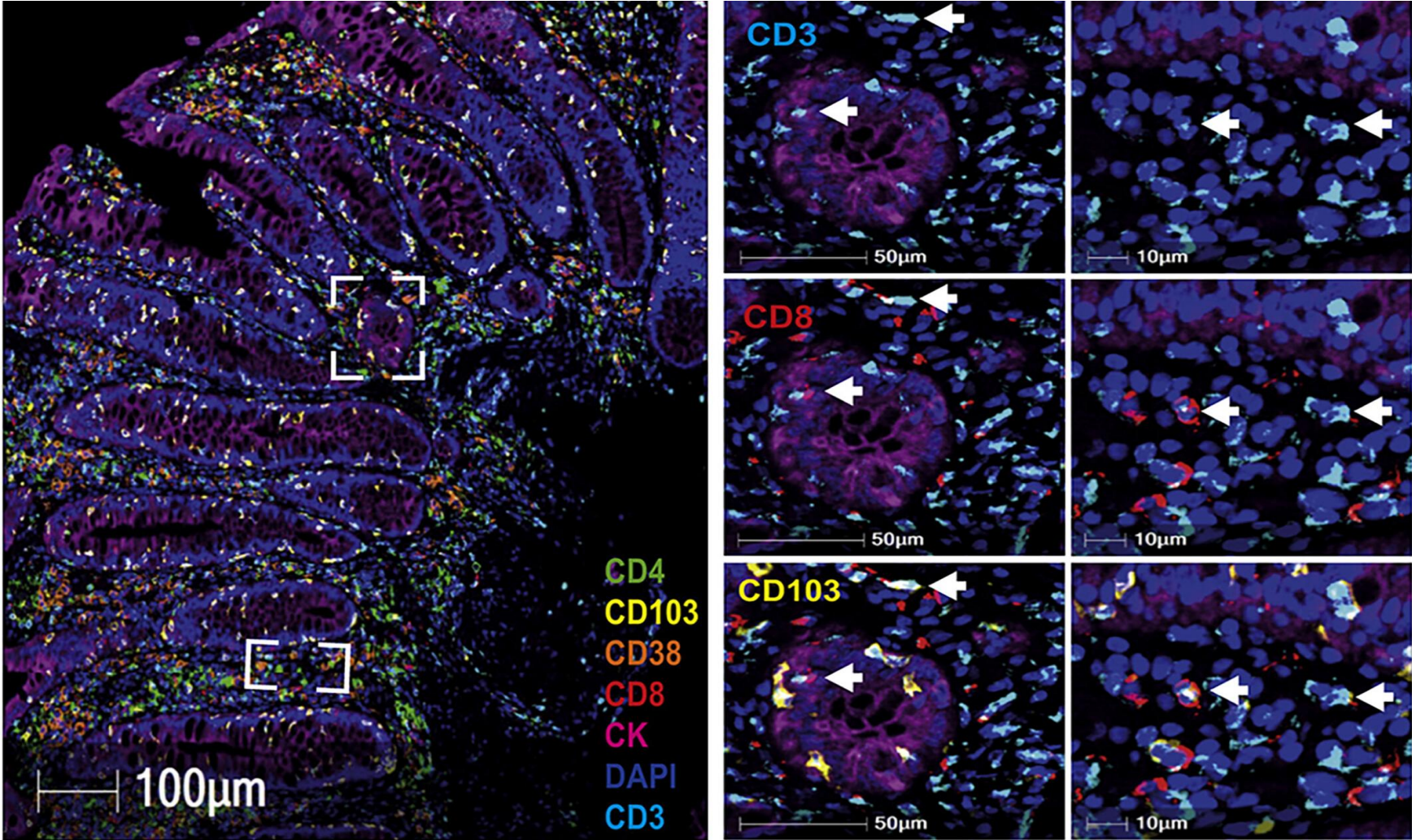
Suppressor T cells are crucial for the maintenance of immunological tolerance. Their major role is to shut down T cell-mediated immunity toward the end of an immune reaction. Suppressor T cells can be called Regulatory T cells Treg.



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Resident immune system



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Immunologic Cell Death-Signaling



Inflammation

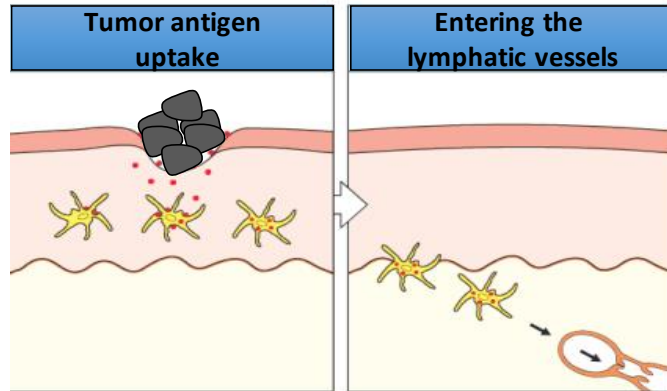
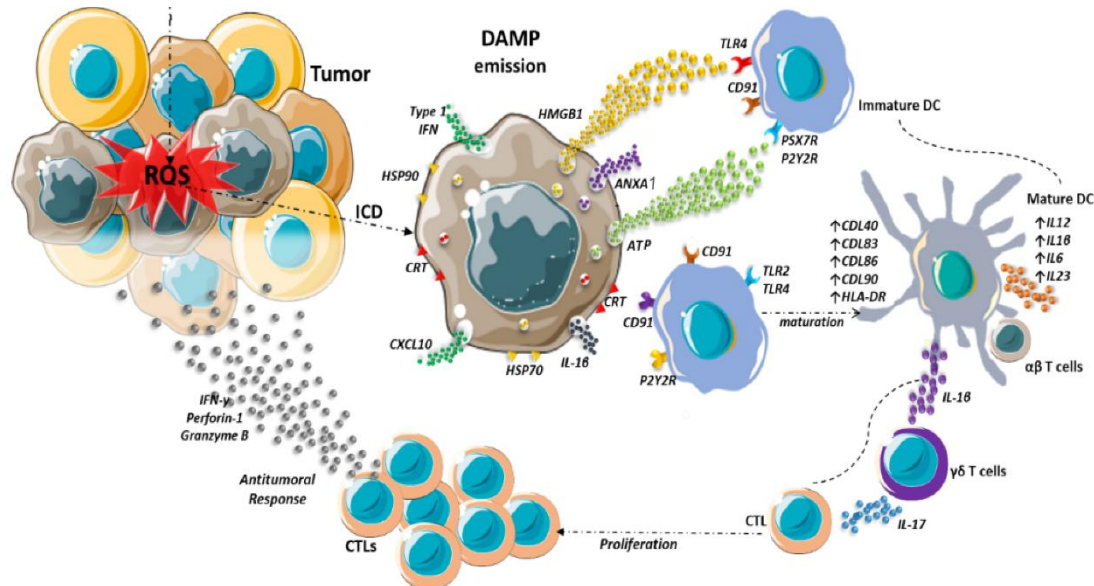


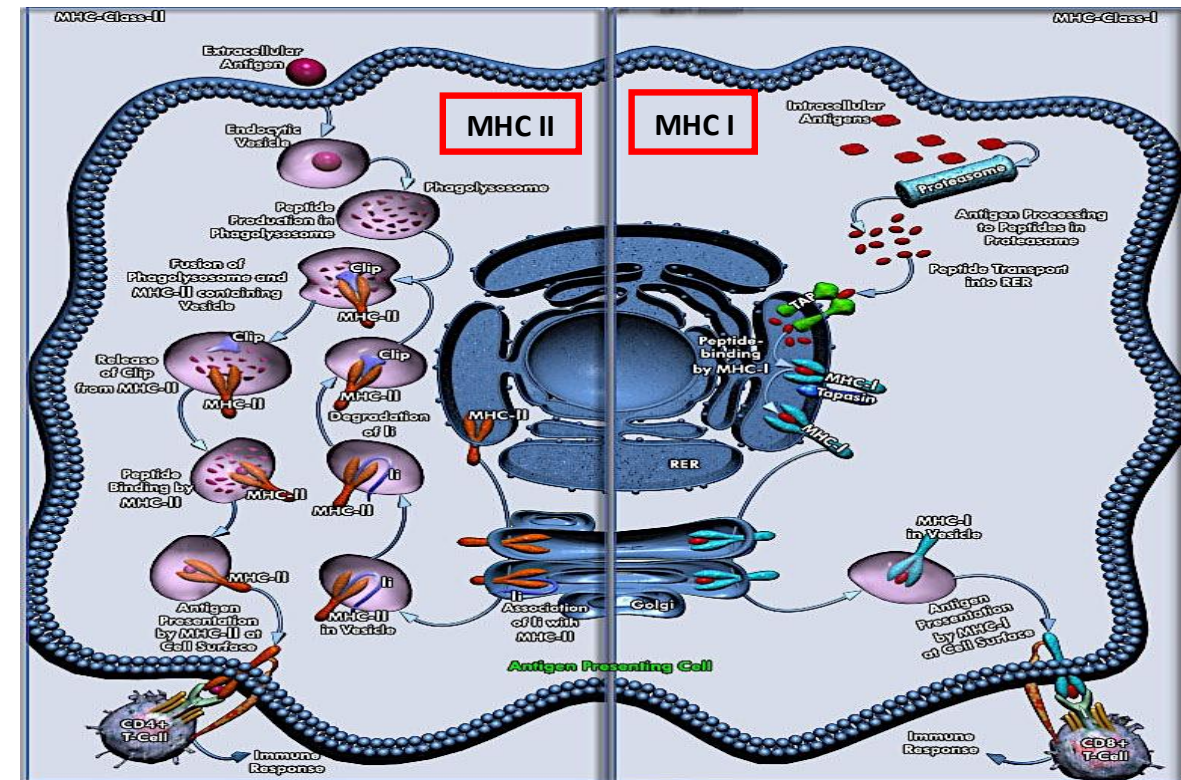
Fig 8.15 part 1 of 2 © 2001 Garland Science

Immature DC



Why are DCs such good APCs?

1. They express both MHC class I and MHC class II
2. They start in the right place (tissues)
3. They end up in the right place (lymph node)
4. Immature DCs are highly efficient at taking up antigens in the tissues.
5. After maturation and migration, mature DCs present these antigens to T cells in the lymph node.
6. They express costimulatory molecules



Comunicazione Cellulare

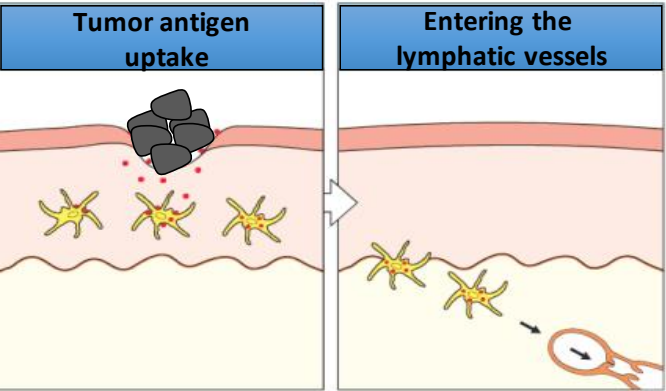


Fig 8.15 part 1 of 2 © 2001 Garland Science

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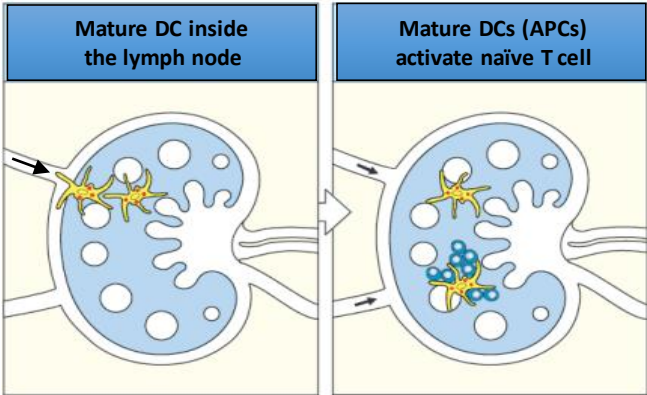
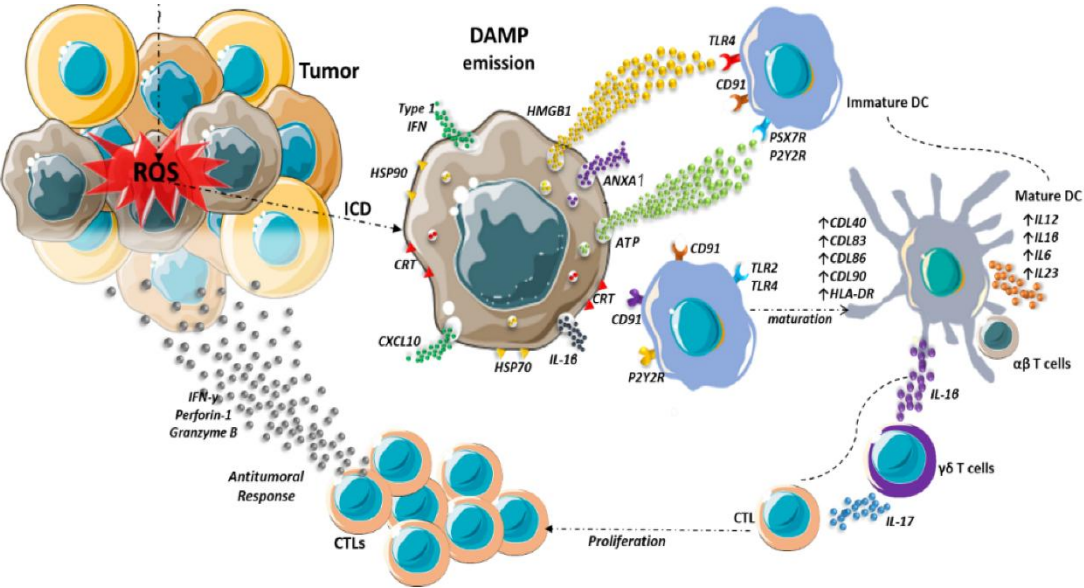


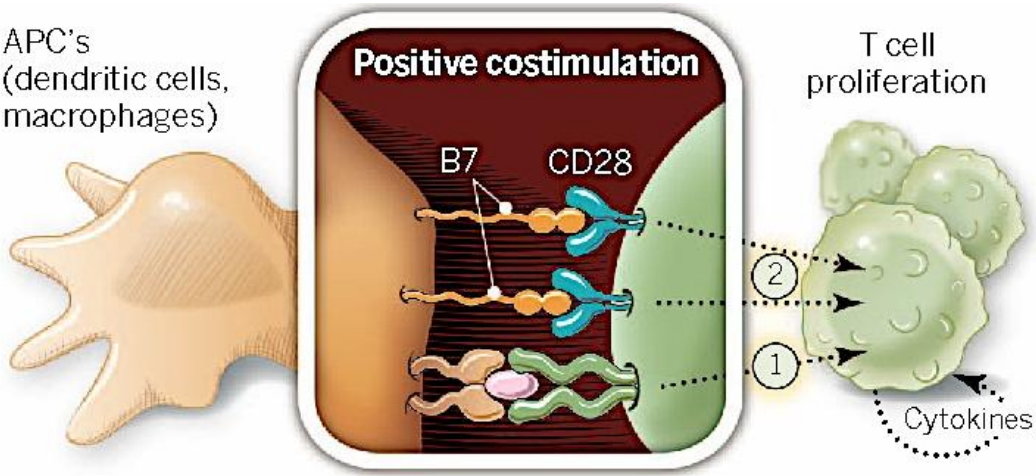
Fig 8.15 part 2 of 2 © 2001 Garland Science

Immature DC



Mature DC

21



Comunicazione Cellulare

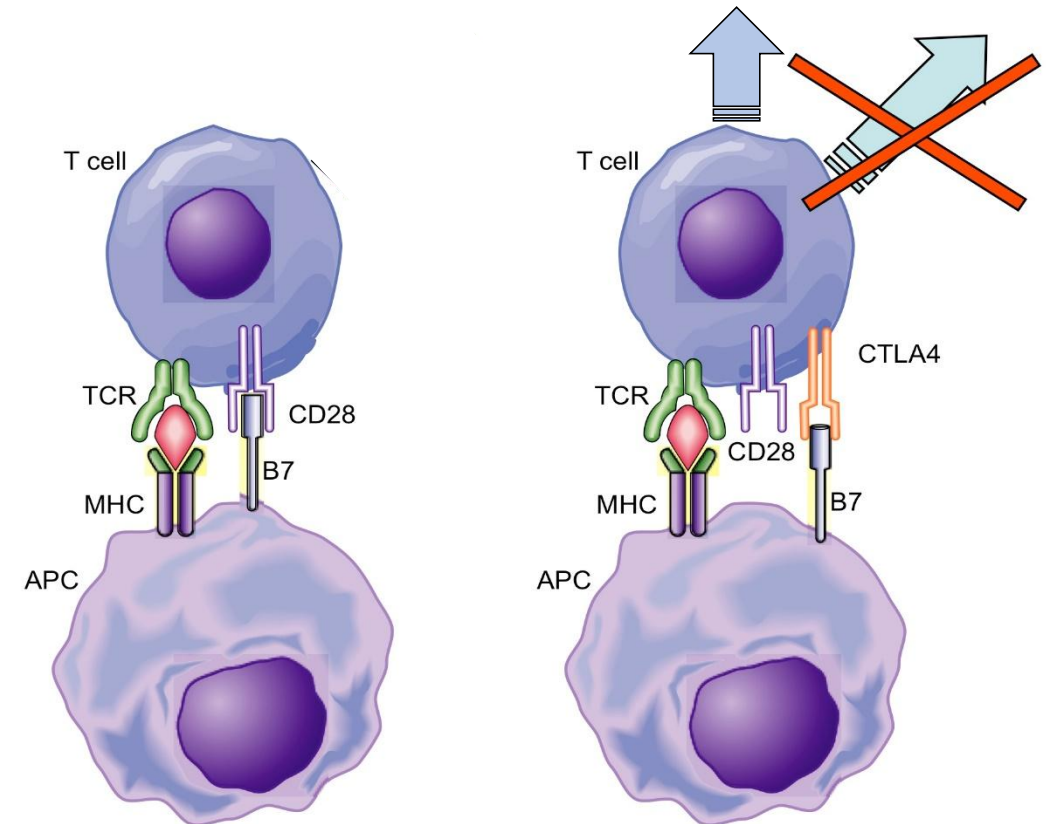
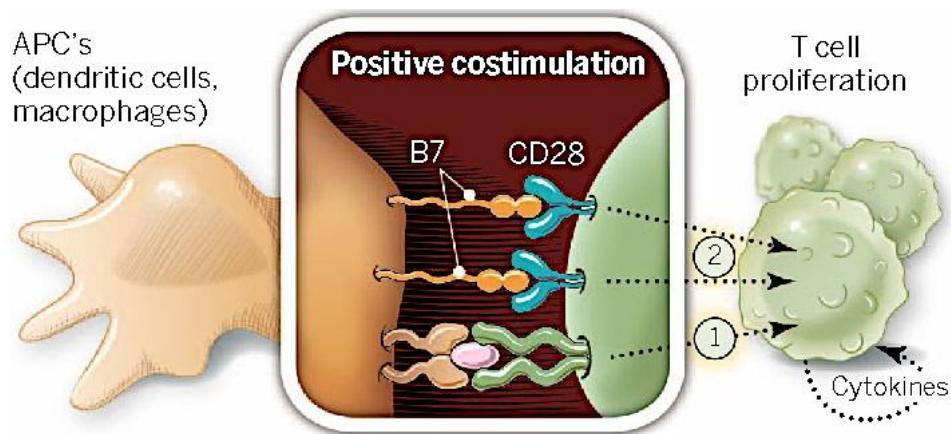
Immunologic Cell Death-Signaling



Inflammation

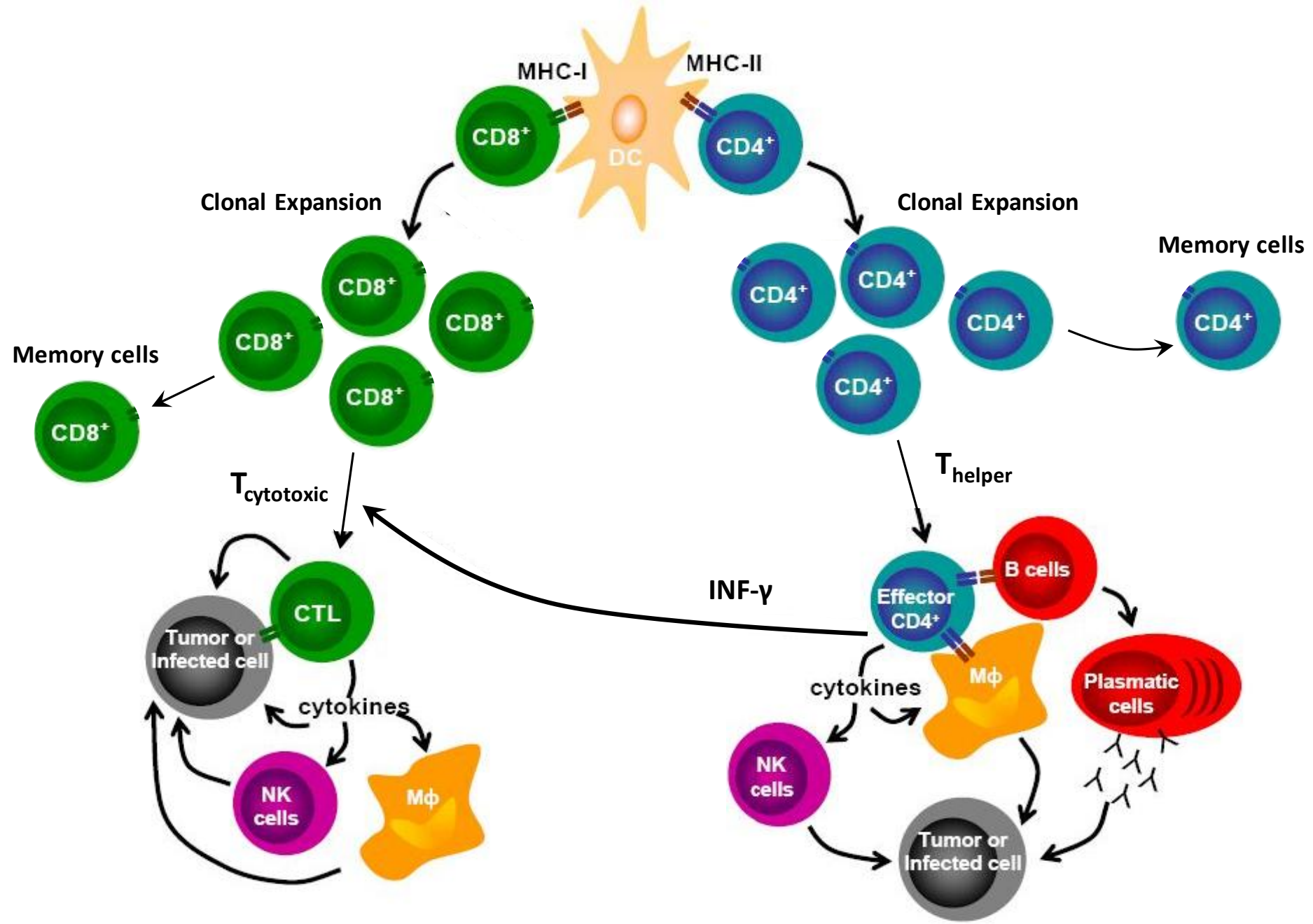
In the case of T cells, the ultimate amplitude and quality of the response, which is initiated through antigen recognition by the T cell receptor (TCR), is regulated by a balance between co-stimulatory and inhibitory signals: **the immune checkpoints**.

Recognition of antigen-MHC complexes by the TCR and costimulatory signals are required for the activation of naïve T cells. These signals are provided by the engagement of CD28 on the T cell surface with B7 molecules (CD80 and CD86) on professional antigen-presenting cells (APC).



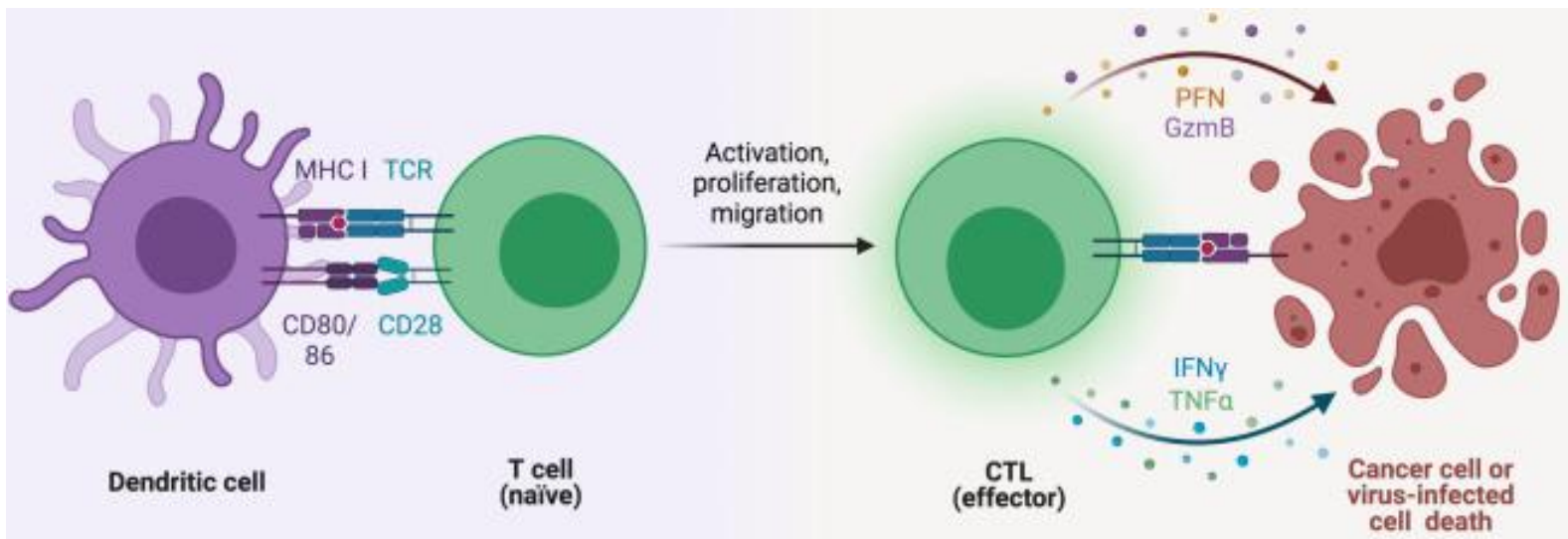
APC, antigen-presenting cell; CTLA-4, cytotoxic T-lymphocyte antigen-4; MHC, major histoc

T/B-cell responses

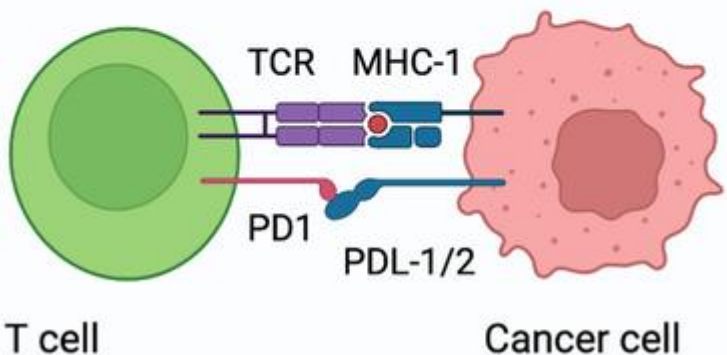


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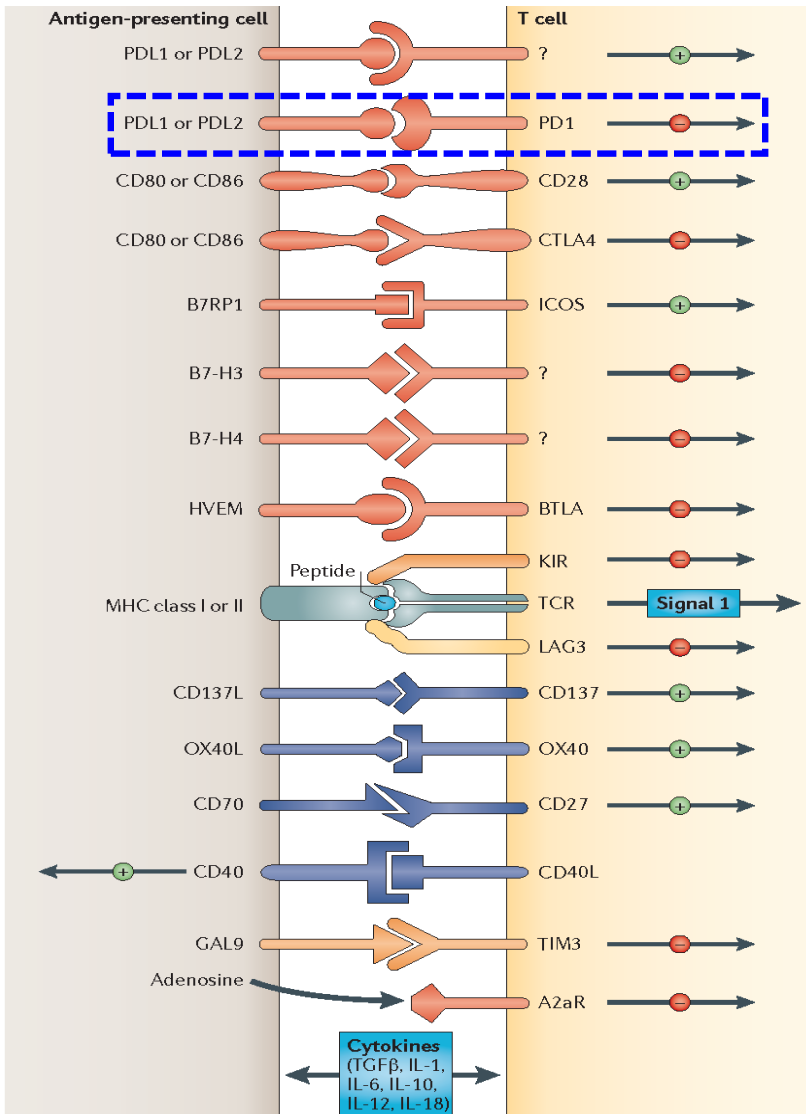
Immune checkpoint: Rules of engagement



PD1-PDL-1/2 complex inhibits tumor cell death



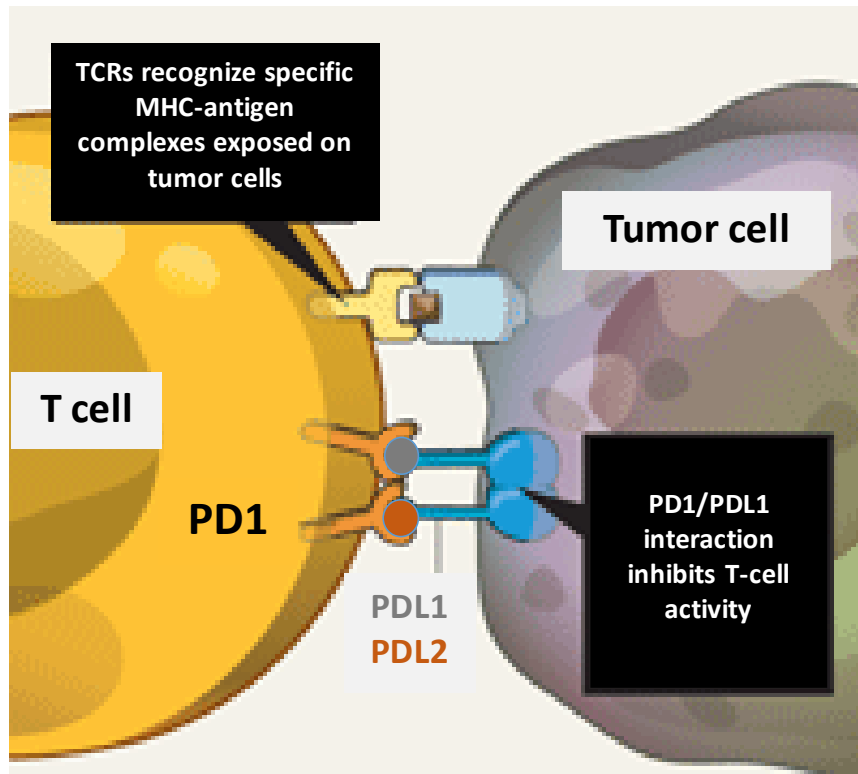
Inhibiting Killing Activity of CD8+ Cytotoxic T-lymphocytes Via Juxtacrine Signals



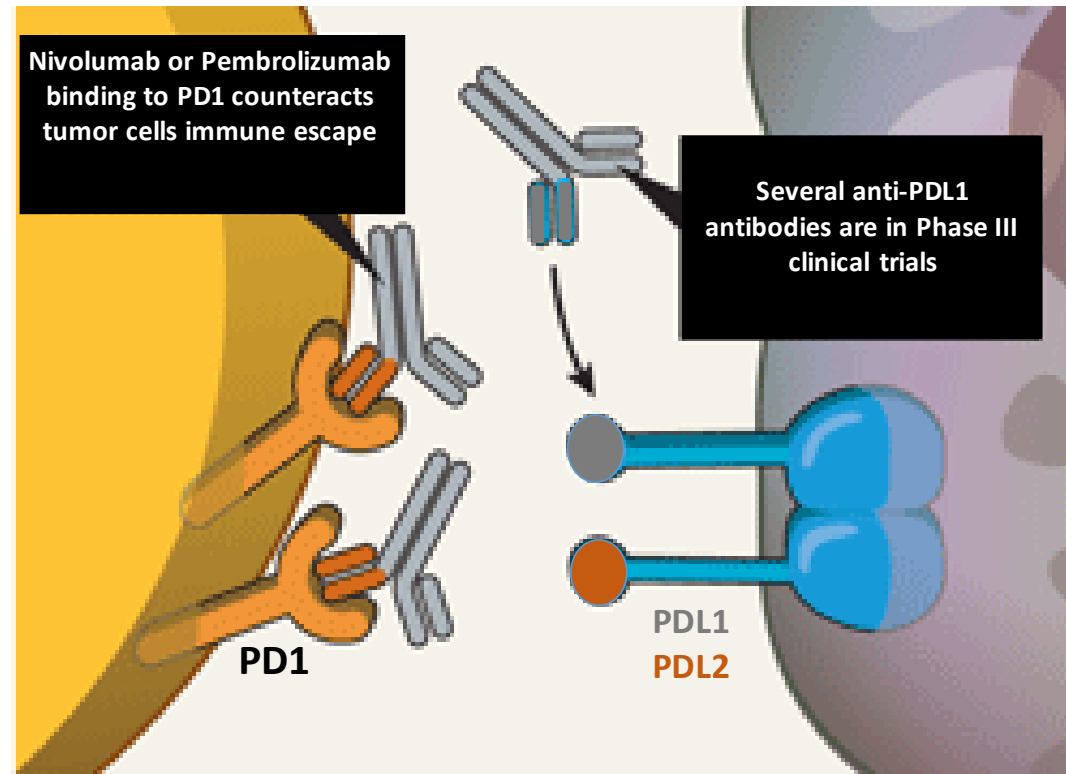
Immune checkpoint modulation for cancer therapy

Targeting PD1/PDL1

PD1/PDL1 mediated immune checkpoint counteracts T cell activities against tumor cells

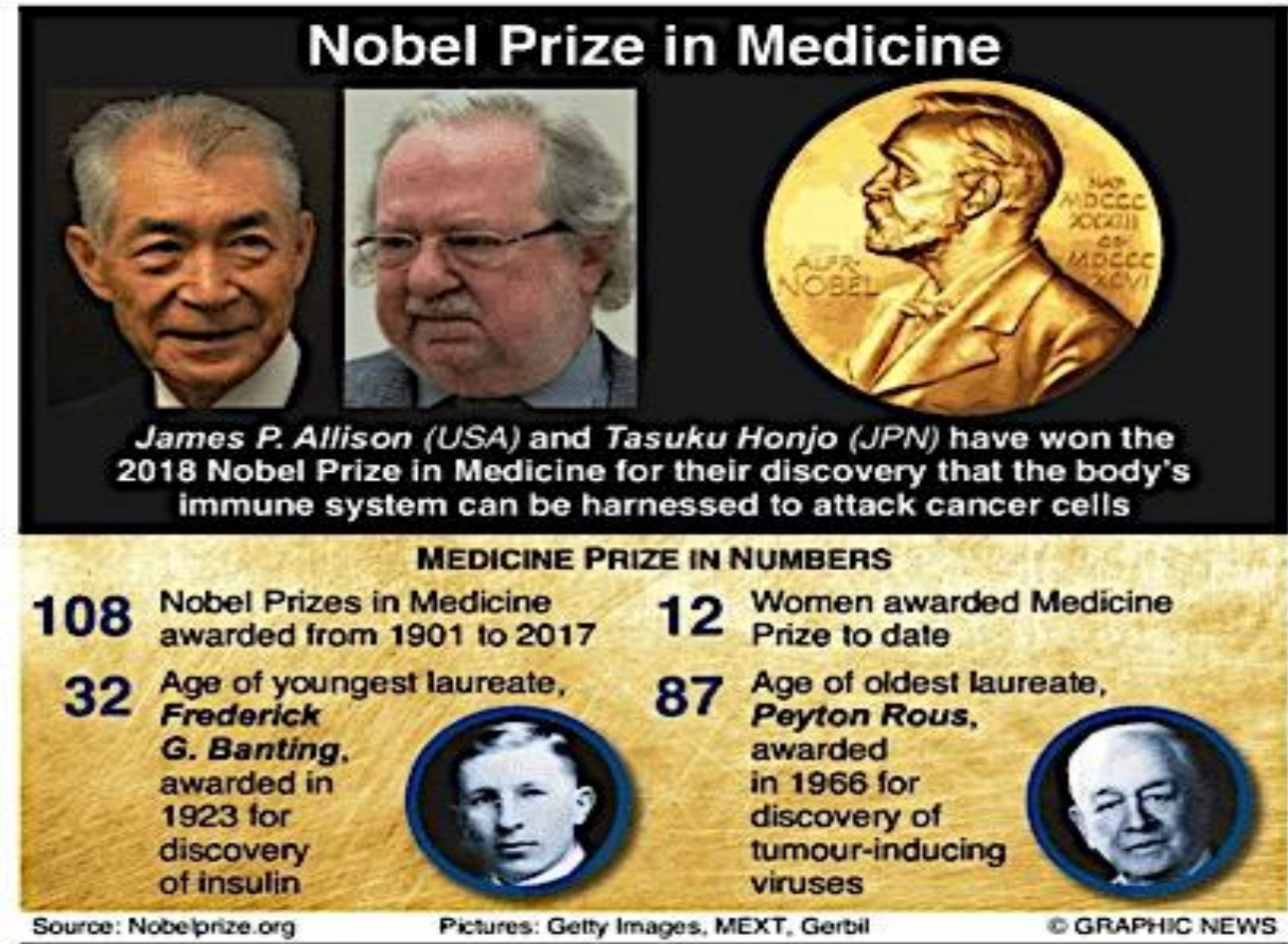
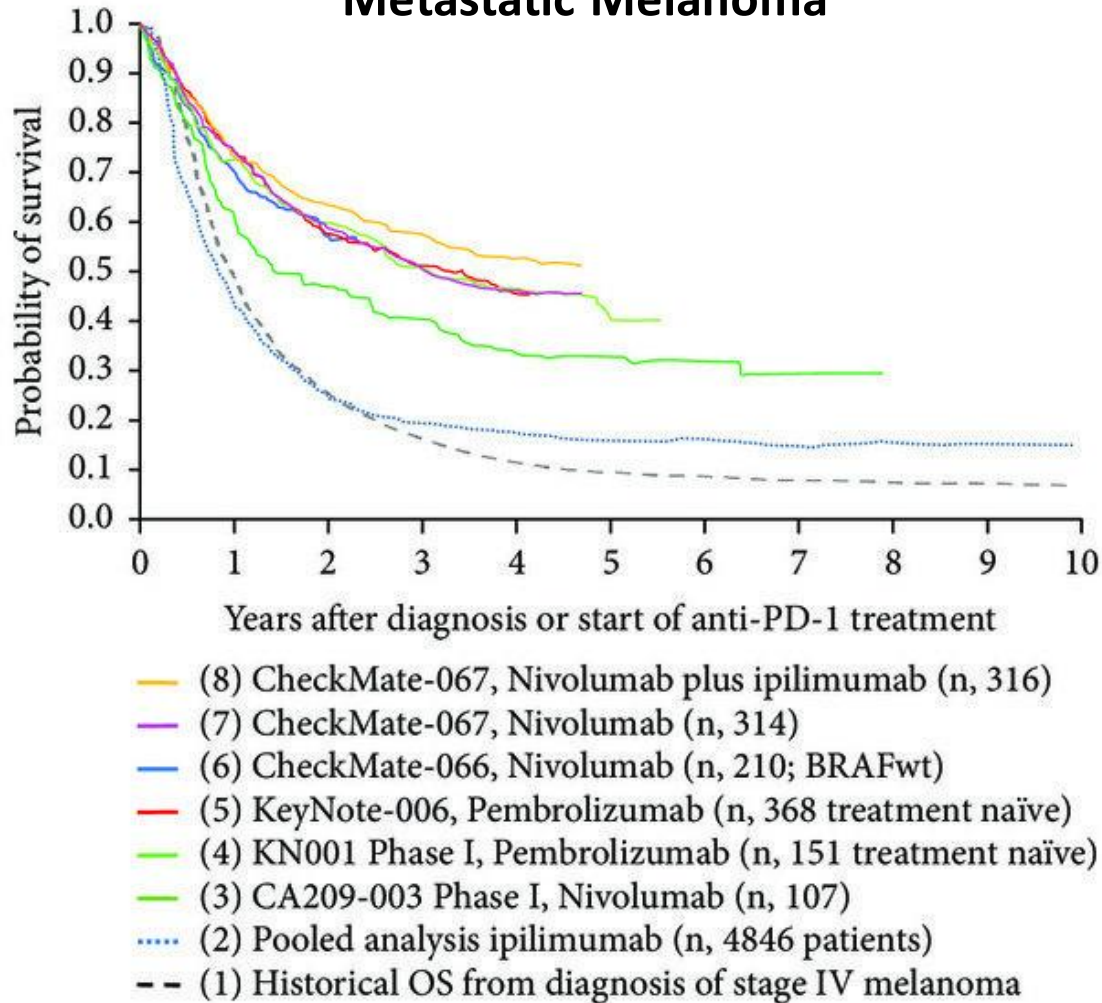


PD1/PDL1 signaling inhibition rescues T cell immune response against tumor cells



Immune checkpoint modulation for cancer therapy

Metastatic Melanoma



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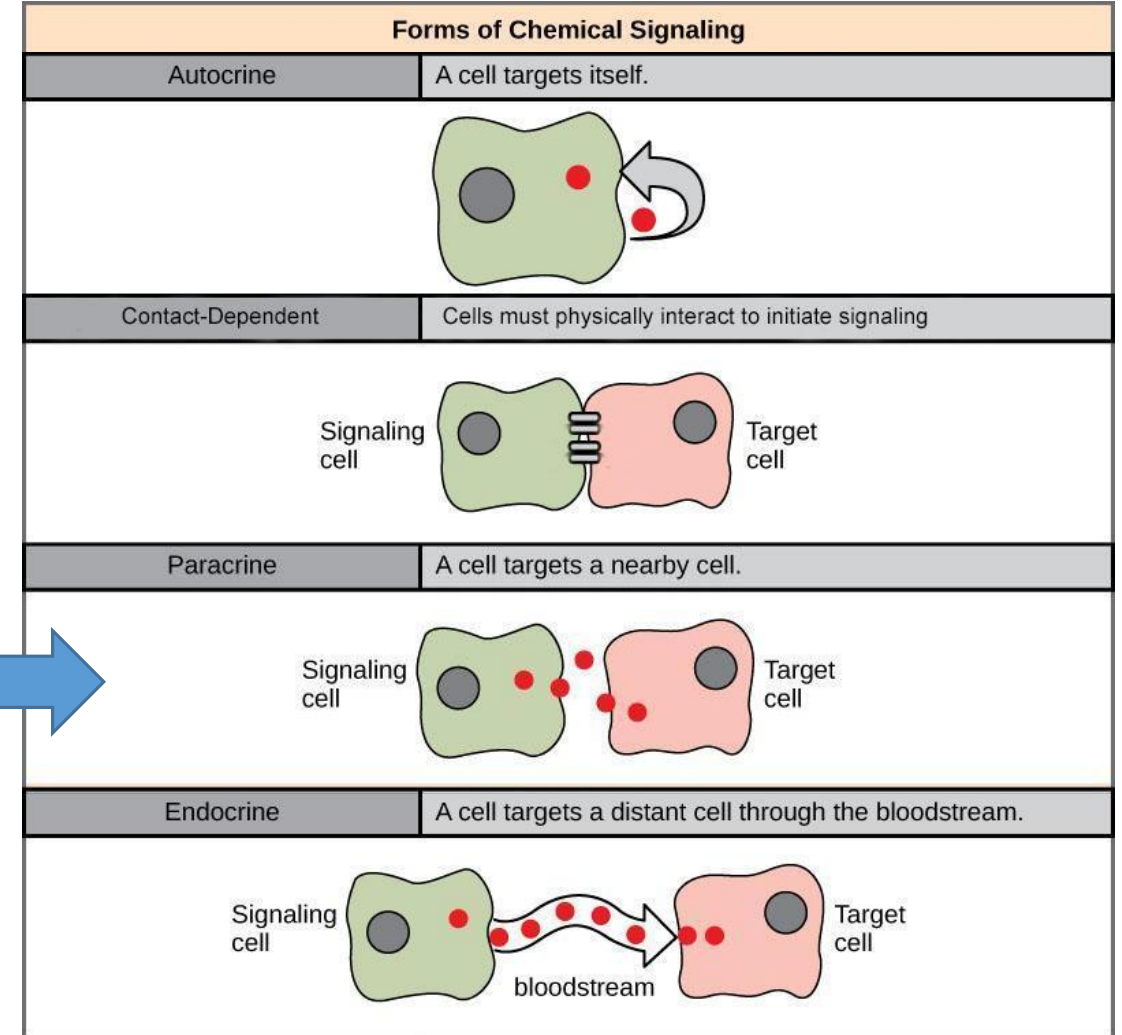
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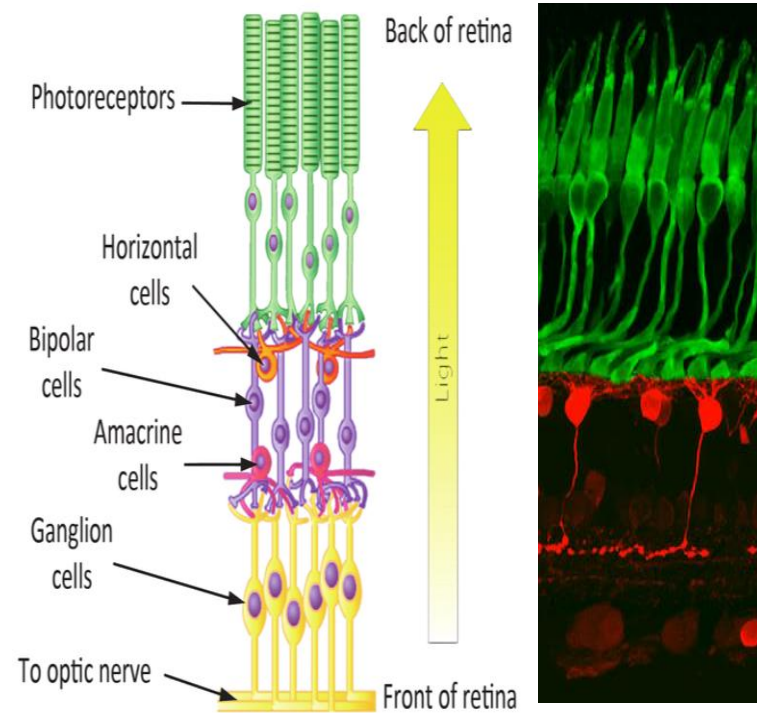
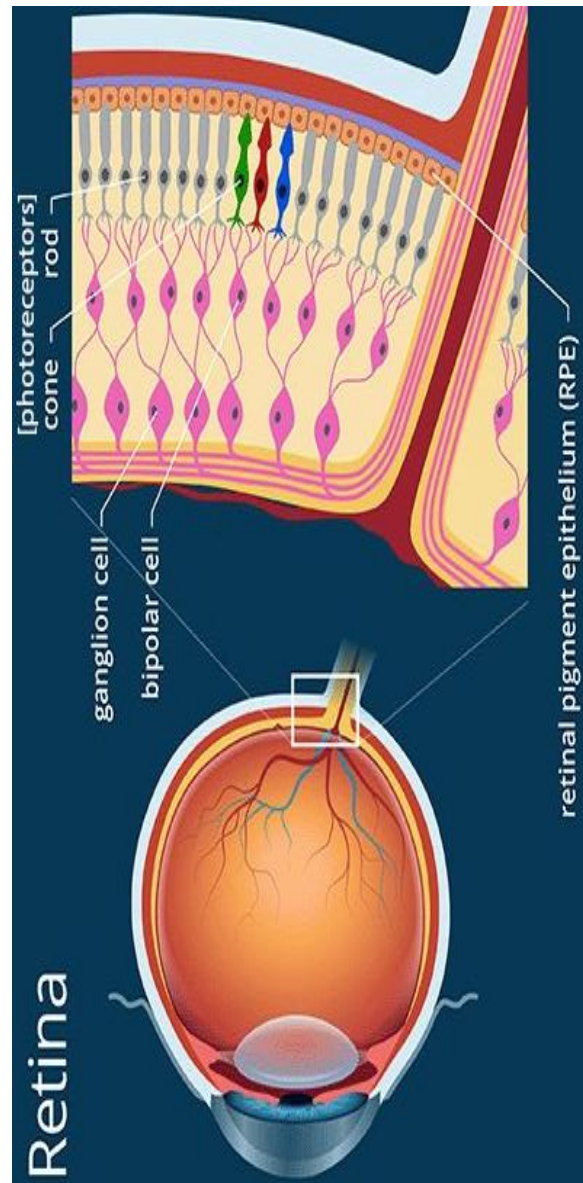
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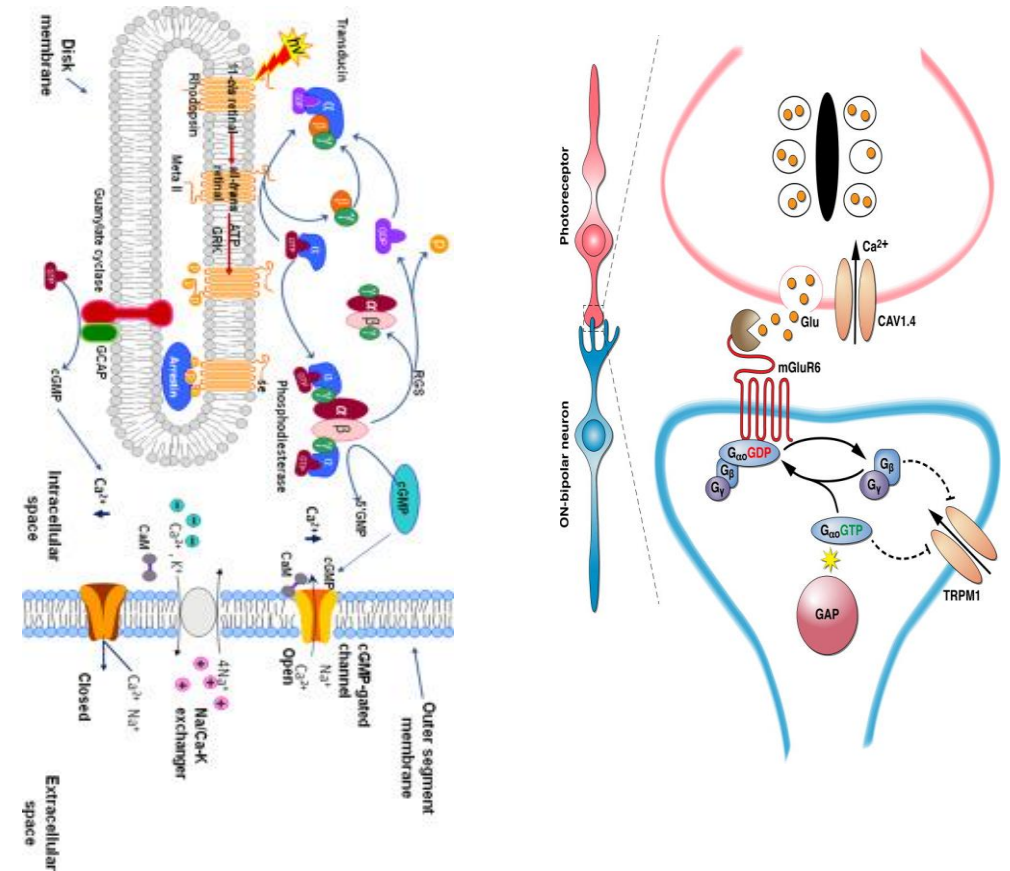


Comunicazione Cellulare: Paracrina



Phototransduction

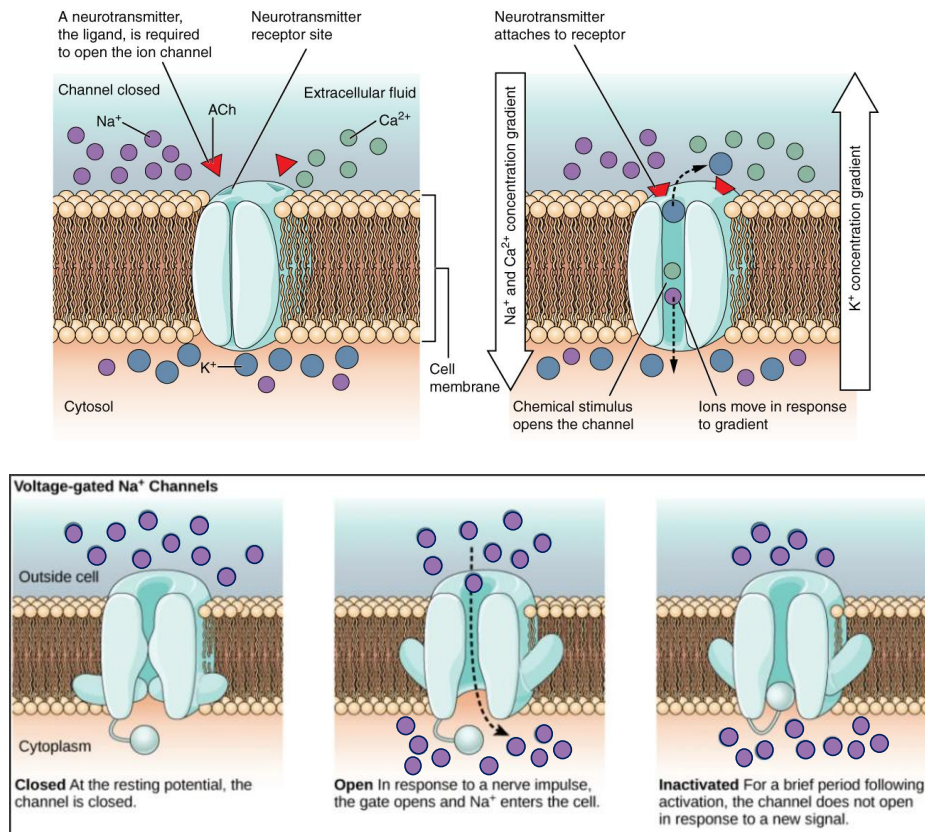
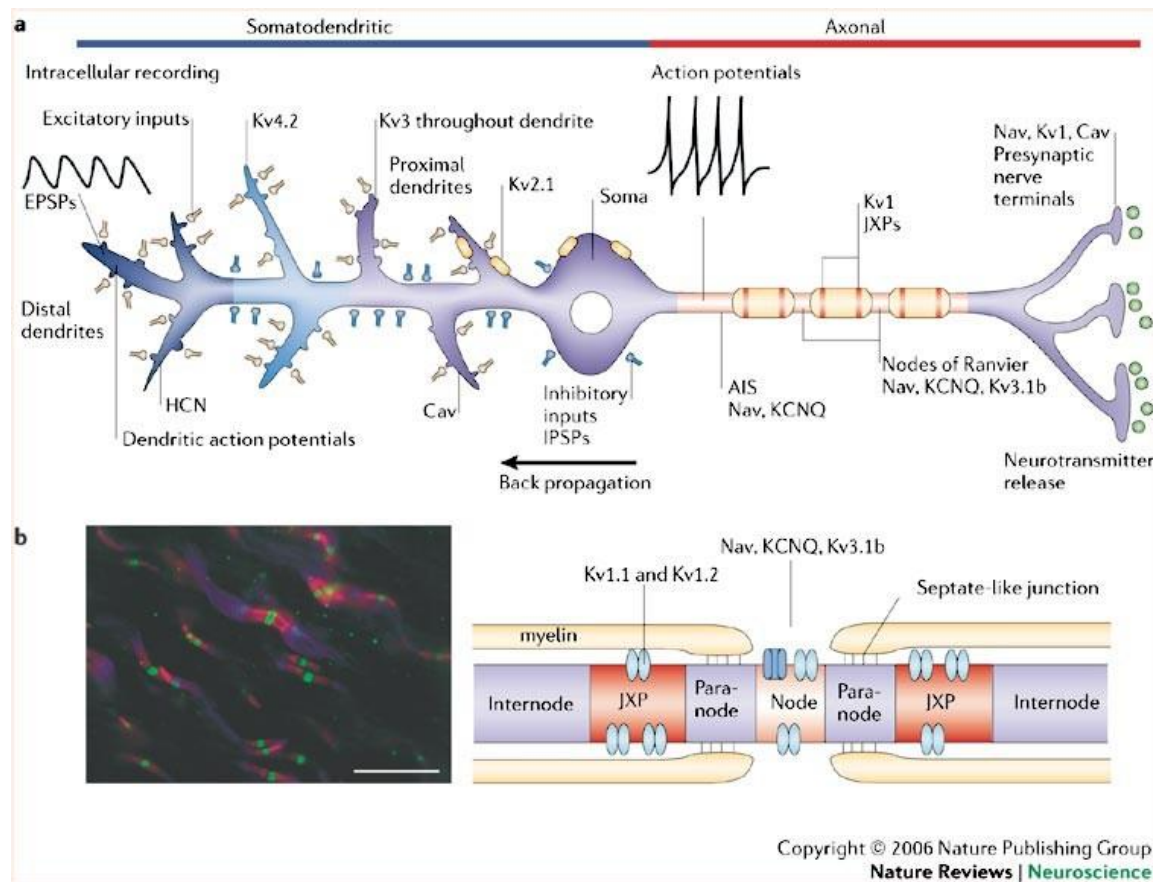
Phototransduction, the process by which a photon of light is changed to an electrical signal, occurs in the photoreceptors. Visual pigments in the photoreceptor outer segment absorb light, initiating the process of vision. A series of biochemical changes follow and the cell hyperpolarizes, which starts an electrical current flow through the retina. The signal passes to bipolar and horizontal cells, some organization and processing occurs, with more organization and processing occurring as the signal is transferred to amacrine and ganglion cells. Once a ganglion cell is activated, its axon carries the message to the brain.



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Ion Channels and Neuronal Communication

Stimulation of neurons by sensory receptor cells partially opens ion channels in the membrane. Sodium diffuses into the cell, shifting that part of the membrane toward a less-negative polarization. If this local potential reaches a critical state called the threshold potential (measuring about $-60/-50$ mV), then voltage-gated sodium channels open and sodium floods that part of the cell, which instantly depolarizes to an action potential of about $+55$ mV. Depolarization activates sodium channels in adjacent parts of the membrane, so that the impulse moves along the fiber.



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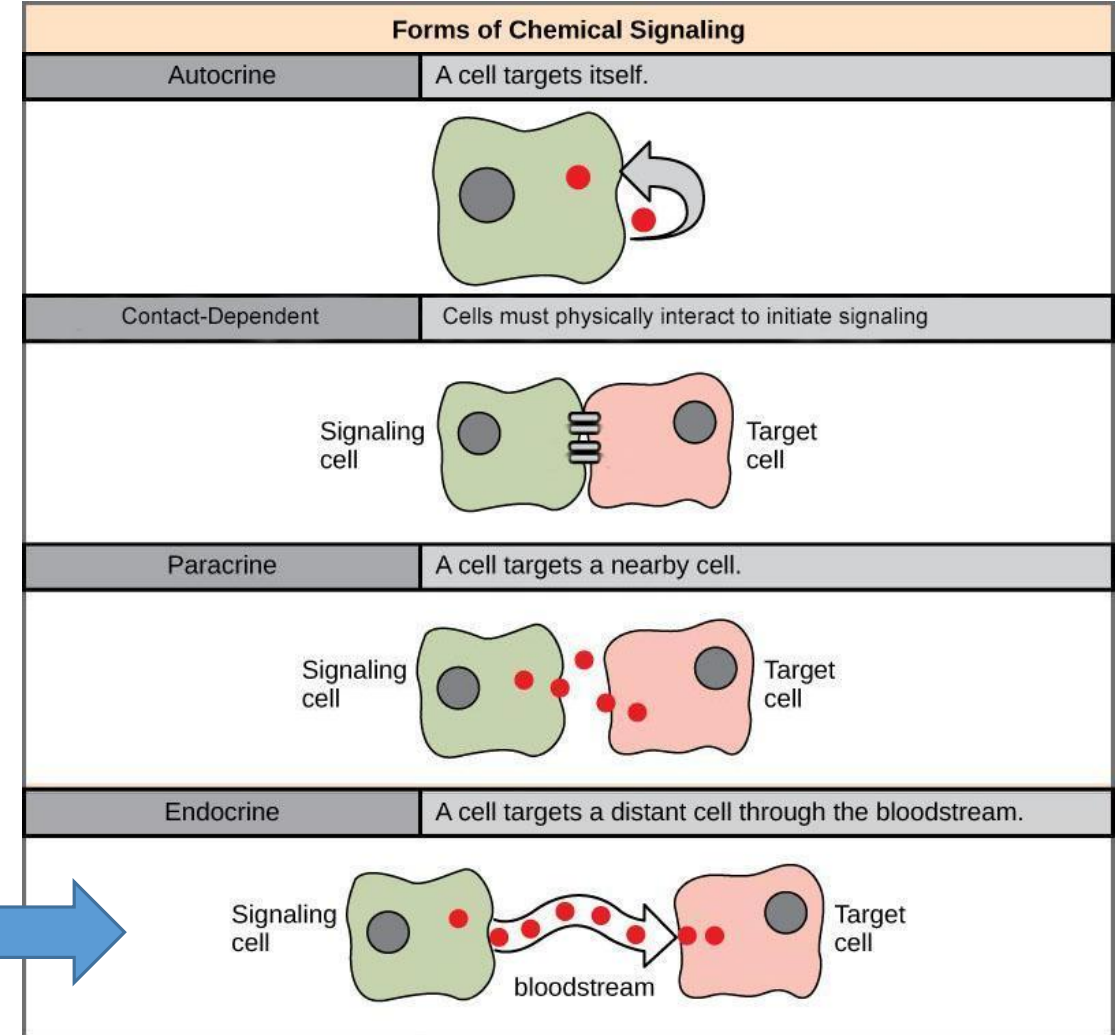
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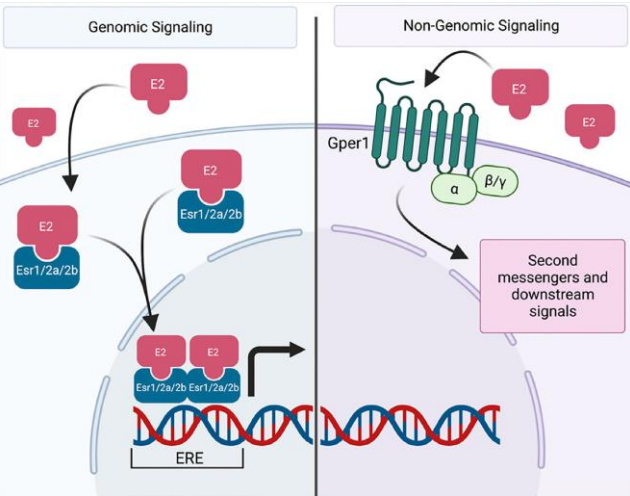
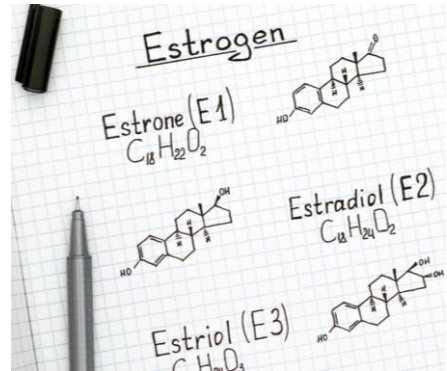
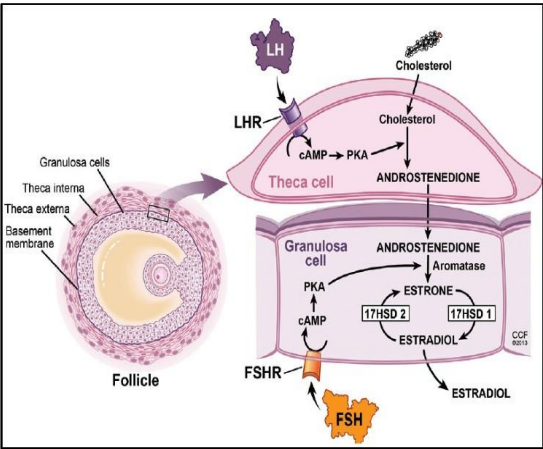
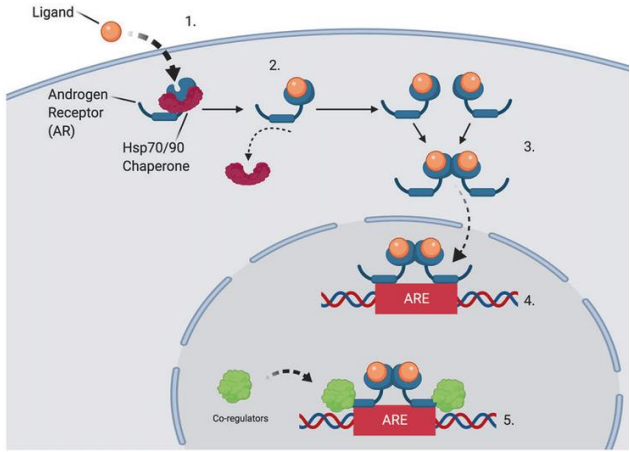
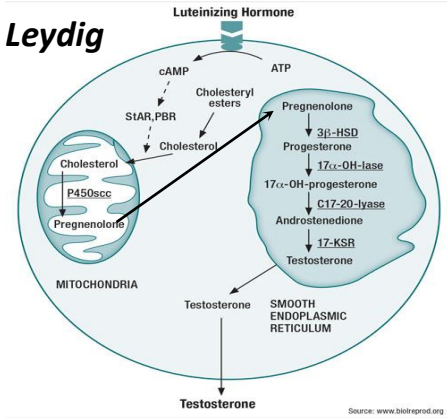
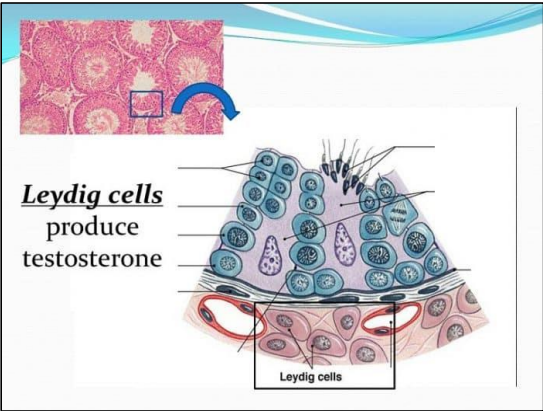
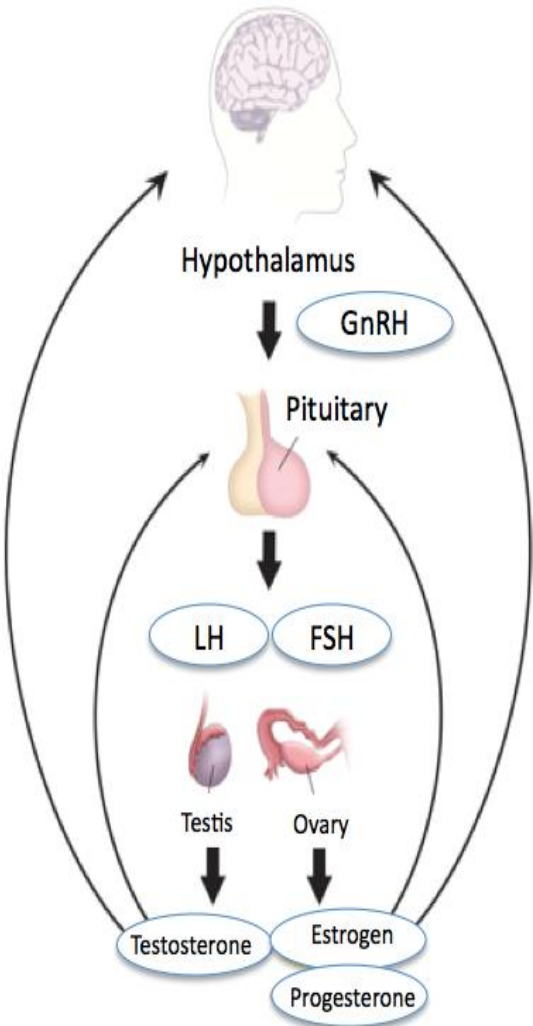


Comunicazione Cellulare: Endocrina

Steroid Hormones

Hormones are substances that function as chemical messengers in the body.

They bind and activate specific receptors that, in turn, regulate cells and tissues physiology.



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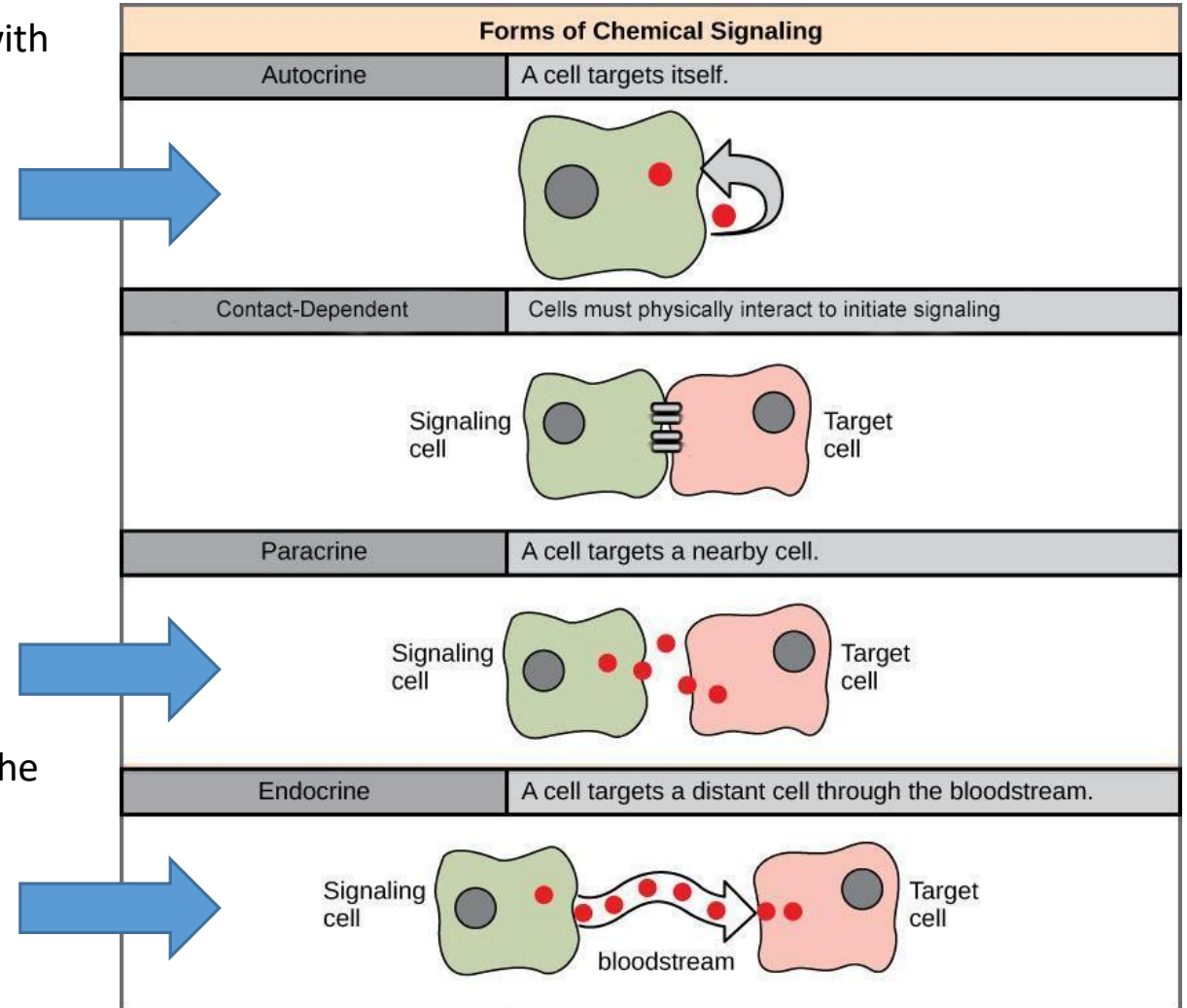
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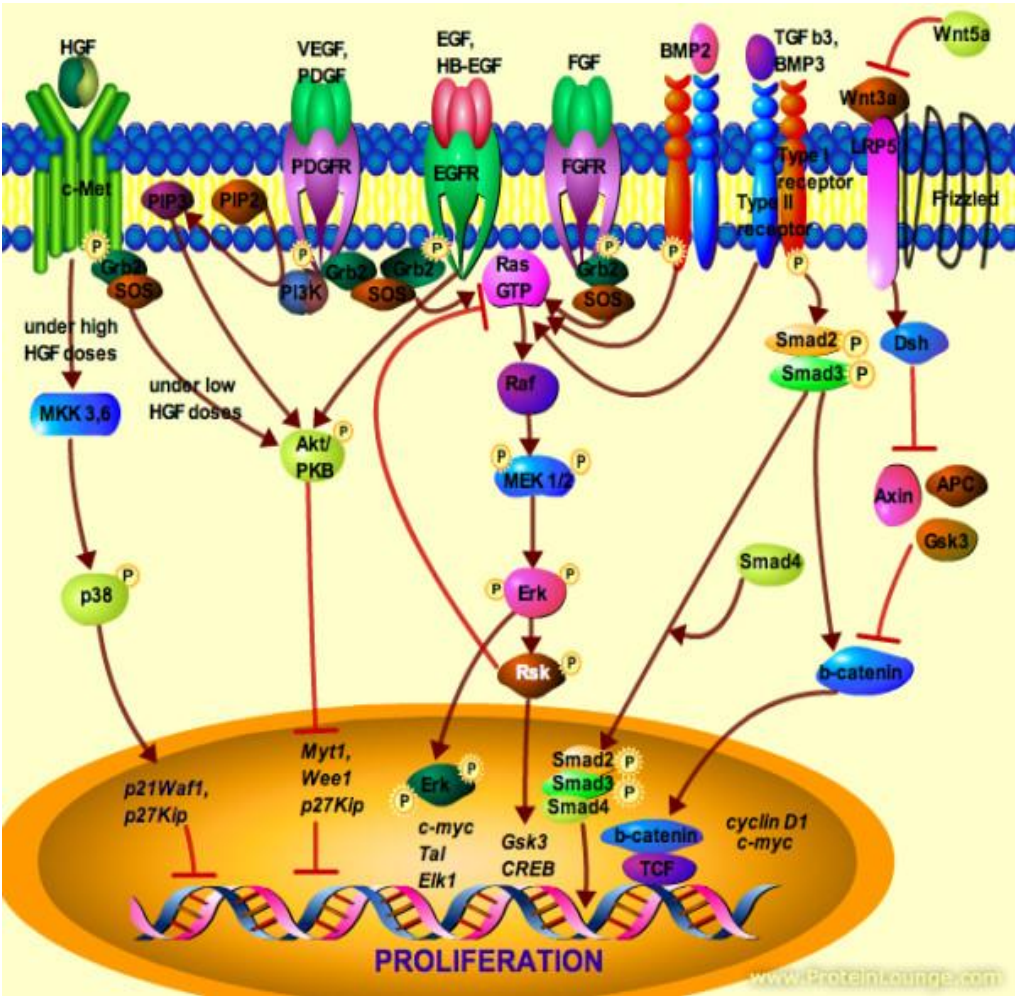
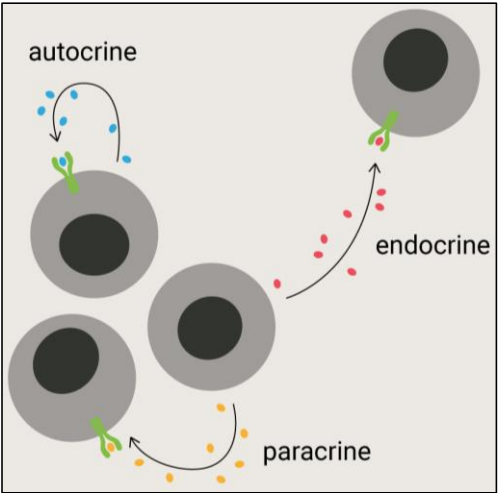


Comunicazione Cellulare

Growth Factors

Growth factors are diffusible signaling proteins that stimulate cell growth, differentiation, survival, inflammation, tissue repair, etc.. They can be secreted by neighboring cells, distant tissues and glands, or even tumor cells themselves.

Growth factor	Major target cells
Interleukins	Various, mainly cells mediating immunity and inflammation
Interferon- γ	Mainly lymphocytes and additional cells mediating immunity (and inflammation)
Colony stimulating factors	Mainly haemopoietic cells
Erythropoietin	Erythroid precursor cells
Thrombopoietin	Mainly megakaryocytes
Neurotrophic factors	Several, but mainly neuronal cell populations
Insulin	Various
Insulin-like growth factors	A very wide range of cells found in various tissue types
Epidermal growth factor	Various, including epithelial and endothelial cells and fibroblasts
Platelet-derived growth factor	Various, including fibroblasts, glial cells and smooth muscle cells
Fibroblast growth factors	Various, including fibroblasts, osteoblasts and vascular endothelial cells
Transforming growth factors- α	Various
Leukaemia inhibitory factor	Mainly various haemopoietic cells



Ligand binding to Receptor

Receptor Activation

Signaling Cascades:

Activation/Inhibition

Relocation

Stability/Degradation

Transcriptional Regulation of Genes